

■ 3.2.3 Pseudorandom Numbers

<code>Random[]</code>	a pseudorandom real between 0 and 1
<code>Random[Real, xmax]</code>	a pseudorandom real between 0 and <i>xmax</i>
<code>Random[Real, {xmin, xmax}]</code>	a pseudorandom real between <i>xmin</i> and <i>xmax</i>
<code>Random[Complex]</code>	a pseudorandom complex number in unit square
<code>Random[Complex, {zmin, zmax}]</code>	a pseudorandom complex number in the rectangle defined by <i>zmin</i> and <i>zmax</i>
<code>Random[type, range, n]</code>	a pseudorandom number with <i>n</i> -digit precision
<code>Random[Integer]</code>	0 or 1 with probability $\frac{1}{2}$
<code>Random[Integer, {imin, imax}]</code>	a pseudorandom integer between <i>imin</i> and <i>imax</i> , inclusive
<code>SeedRandom[]</code>	reseed the pseudorandom generator, with the time of day
<code>SeedRandom[s]</code>	reseed with the integer <i>s</i>

Pseudorandom number generation.

This gives a list of three pseudorandom numbers.

```
In[1]:= Table[Random[ ], {3}]
Out[1]= {0.353297, 0.0205034, 0.117374}
```

Here is a 30-digit pseudorandom real number in the range 0 to 1.

```
In[2]:= Random[Real, {0, 1}, 30]
Out[2]= 0.756200220413389091346356107147
```

This gives a list of eight pseudorandom integers between 100 and 200 (inclusive).

```
In[3]:= Table[Random[Integer, {100, 200}], {8}]
Out[3]= {127, 195, 196, 113, 187, 117, 155, 154}
```

If you call `Random[ ]` repeatedly, you should get a “typical” sequence of numbers, with no particular pattern. There are many ways to use such numbers.

One common way to use pseudorandom numbers is in making numerical tests of hypotheses. For example, if you believe that two symbolic expressions are mathematically equal, you can test this by plugging in “typical” numerical values for symbolic parameters, and then comparing the numerical results. (If you do this, you should be careful about functions of complex variables that may not have unique

values.)

Here is a symbolic equation.

```
In[4]:= Sin[Cos[x]] == Cos[Sin[x]]
Out[4]= Sin[Cos[x]] == Cos[Sin[x]]
```

Substituting in a random numerical value shows that the equation is not always `True`.

```
In[5]:= % /. x -> Random[]
Out[5]= False
```

Other common uses of pseudorandom numbers include simulating probabilistic processes, and sampling large spaces of possibilities. The pseudorandom numbers that *Mathematica* generates are always uniformly distributed over the range you specify.

`Random` is unlike any other *Mathematica* function in that every time you call it, you potentially get a different result. If you use `Random` in a calculation, therefore, you may get different answers on different occasions.

The sequences that you get from `Random[]` are not in most senses “truly random”, although they should be “random enough” for practical purposes. The sequences are in fact produced by applying a definite mathematical algorithm, starting from a particular “seed”. If you give the same seed, then you get the same sequence.

When *Mathematica* starts up, it takes the time of day (measured in small fractions of a second) as the seed for the pseudorandom number generator. Two different *Mathematica* sessions will therefore almost always give different sequences of pseudorandom numbers.

If you want to make sure that you always get the same sequence of pseudorandom numbers, you can explicitly give a seed for the pseudorandom generator, using `SeedRandom`.

This re-seeds the pseudorandom generator.

```
In[6]:= SeedRandom[143]
```

Here are three pseudorandom numbers.

```
In[7]:= Table[Random[], {3}]
Out[7]= {0.353297, 0.0205034, 0.117374}
```

If you re-seed the pseudorandom generator with the same seed, you get the same sequence of pseudorandom numbers.

```
In[8]:= SeedRandom[143]; Table[Random[], {3}]
Out[8]= {0.353297, 0.0205034, 0.117374}
```